

Behavioural-to-Video-Ephys Temporal Alignment and Neural Circuit Characterisation

1st Cycle Integrated Project in Engineering Physics

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1 Introduction and Project Scope

This report consolidates three sequential phases of a single alignment problem: relating events in a behavioural log (a head-fixed joystick-pull task delivering water reward) to two independently clocked recording systems — a behaviour-tracking video camera and a Neuropixels 1.0 electrophysiology probe (ChocolateGroup cohort, animals Ferrero Rocher and Milka, recorded August 2023).

- **Phase 0 (Video alignment).** An ensemble video-tracking method (PC1 trace + template matching + spatial stability) detected a physical event in raw video and matched it to behaviour-log timestamps, validating that confidence-weighted ensemble detection could produce a usable video-behaviour warp.
- **Phase 1 (Continuous-feature ephys alignment).** With no functional hardware sync channel, over 440 continuous electrophysiological features were engineered and audited as candidate alignment anchors. None proved specific enough to reward delivery to support precise alignment.
- **Phase 2 (Spike-sorted anchor alignment).** Individual well-isolated units from Kilosort4 were used as biological clocks: a single reward-responsive neuron, matched per trial, supports a linear clock warp with sub-millisecond to ~ 20 ms precision. This phase succeeded and is the foundation for the neural circuit analysis presented in Sec. 4.

Sections 2–3 summarise the two preliminary phases; Sec. 4 covers the successful spike-sorted phase in full detail, as it is both the methodological core of the project and the basis for all downstream scientific conclusions. Section 5 synthesises across phases.

2 Phase 0: Video–Behaviour Alignment

Session 4_Milka R1 was processed with `video_alignment_run_ready.py` (180,745 frames at frame step 10), extracting per-frame PC1 and template-match scores and combining them into a confidence-weighted ensemble detector. A 15-configuration grid search over `min_conf` $\in \{0.2, \dots, 0.5\}$ and `pc1_weight` $\in \{0.3, 0.45, 0.6\}$ selected `min_conf` = 0.3, `pc1_weight` = 0.6 as the best configuration (MAE = 3.09 s, RMSE = 3.68 s). All 138 trials in the session were successfully aligned (100%), with a fitted warp of $t^{\text{video}} = 1.0021 t^{\text{bhv}} + 1.15$ s and a total clock drift of ~ 4.8 s over the 2300 s session (Table 1).

Metric	Value
Warp stretch / offset	1.0021 / 1.15 s
Clock drift (total)	~ 4.8 s over 2300 s
Trials aligned	138 / 138 (100%)
Median confidence	~ 0.80

Tab. 1: Video–behaviour warp summary, Milka R1.

Known limitation, carried forward. The warp fit’s initial offset bound (± 5 s) was hit during fitting, producing a suboptimal initial warp and an inflated raw RMSE of 151 s in the sweep output (the 3.68 s figure above reflects the corrected fit). This bound was widened to ± 100 s for subsequent sessions. This phase established that ensemble confidence-weighted video detection is viable at second-scale precision, but is not precise enough on its own for trial-level neural alignment — motivating the ephys-specific work in Phases 1–2.

3 Phase 1: Continuous-Feature Ephys Alignment

3.1 Problem and Approach

The Neuropixels hardware sync channel (AP channel 384) was flat across all 89 binary files in the sessions of primary interest — confirmed directly via `np.unique(sync) -> [64]` (a floating baseline, no TTL). With no hardware ground truth, the pipeline (`ephys_alignment_fusion.py`) searched for a continuous neural or physical signal time-locked to reward delivery: solenoid electromagnetic artefact (raw, blocked, derivative variants), MUA envelope, LFP band power (delta/theta/beta/gamma, per depth block, PCA), phase-amplitude coupling, reward-modulation index, common-average-referenced (CAR) MUA, and spike-density function — in total 440–520 features per session.

3.2 Result: Negative

No feature achieved usable specificity. The best F1 score across all features was 0.118 (`pac_block_8`); a usable alignment anchor would need $F1 > 0.5$. The structural problem: every feature responds to general movement, licking, and arousal in addition to (or instead of) reward delivery, so candidate peaks vastly outnumber true reward events — e.g. `solenoid_block` detected 13,055 peaks against 336 rewards (Fig. 1). ERP traces using the resulting 167 ms-RMSE alignment were flat for nearly all features (Fig. 2 shows the one partial exception, with visible but small trial-locked structure), confirming the alignment was too coarse to resolve genuine neural responses.

Method	RMSE (ms)	Notes
LFP deflection (best continuous feature)	144	Still unusable
All 440+ features fused	179	No improvement
Spike-sorted anchor (Phase 2)	10.75	15-fold improvement

Tab. 2: Alignment quality, best continuous methods vs. spike-sorted anchor.

A survey of all 89 AP files found a functional hardware TTL signal (`[0, 64]` sync values) in a minority of sessions (Milka S2/S3, Ferrero S2 imec1), establishing ground-truth sessions usable for later independent validation (Sec. 4.2). The conclusion of this phase: no continuous feature is specific enough to reward delivery for precise alignment; individual spike-sorted units, available in the same Kilosort4 output, were the natural next candidate.

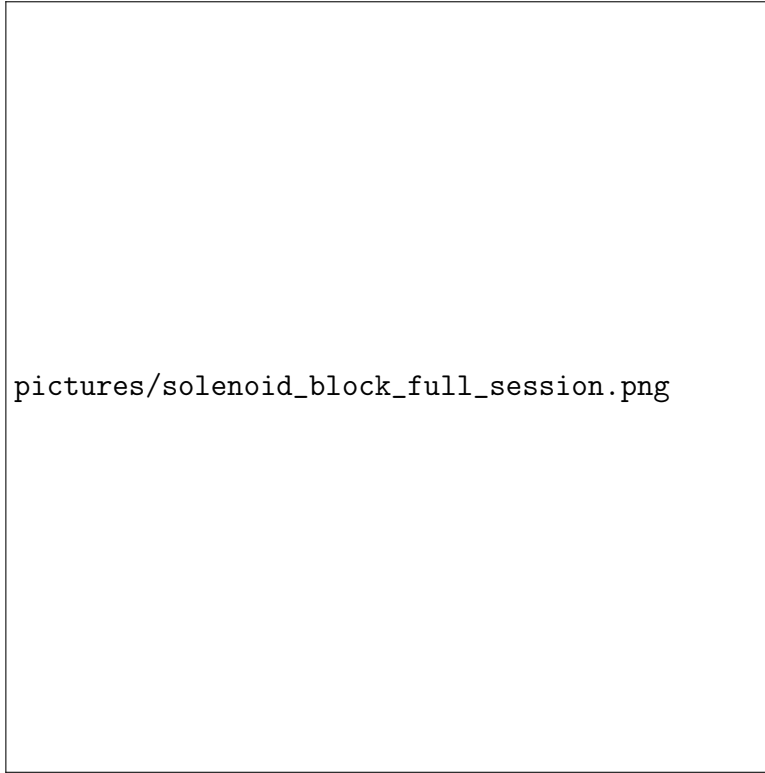


Fig. 1: Feature audit, `solenoid_block`, full session. High hit rate (0.96) but equally high false-peak rate (0.95, $n = 13,055$ peaks vs. 336 rewards) and 178 ms median timing error confirm the feature is sensitive but not specific to reward delivery.

4 Phase 2: Spike-Sorted Anchor Alignment and Neural Circuit Analysis

4.1 Data and Method

Eight probe recordings were analysed (Table 3), all Neuropixels 1.0 (384 channels, 30 kHz), spike-sorted with Kilosort4 (IBL sorter for some Milka imec1 sessions) and manually curated in Phy.

`anchor_unit_hunter.py` ranked good units by $\text{score} = \text{FR}_{\text{peak}} - \text{FR}_{\text{baseline}}$ in 10 ms bins over ± 500 ms around reward events, filtered to $\geq 60\%$ trial reliability. `spike_sync_generator.py` then, per session: zero-anchors the behaviour log to ephys start, matches each reward event to the nearest anchor-unit spike within a ± 50 ms window of the unit’s peak latency, rejects outliers via $3\text{-}\sigma$ iterative clipping, and fits a least-squares linear warp $t^{\text{ephys}} = s \cdot t^{\text{bhv}} + \delta$. For Milka R2 imec0 (single-anchor match rate only 68%), three units were fused via weighted-median voting (`multi_anchor_sync.py`), improving RMSE from 2.14 ms (best single unit) to 1.62 ms.

4.2 Independent Validation Against Hardware Sync

Where Phase 1’s sync-channel survey identified a functional hardware TTL signal, `validate_spike_sync` independently checked the spike-anchor warp against ground truth. For Ferrero S1 imec0, the hardware warp and spike warp agreed to within 6.98 ms RMSE on predicted event time across

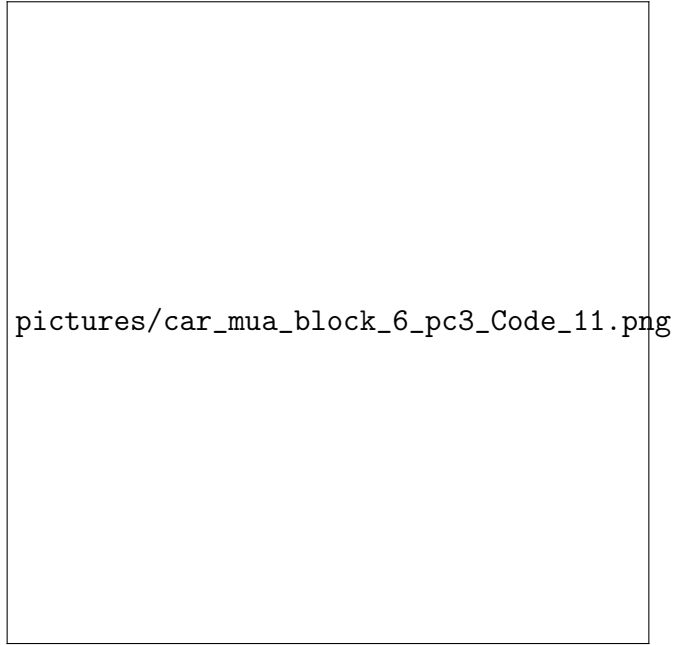


Fig. 2: Trial-resolved ERP, `car_mua_block_6_pc3`, Pull event ($n = 246$ trials) — the closest any continuous feature came to visible trial-locked structure, still too weak relative to trial-to-trial variability to serve as an alignment anchor.

Animal	Session	Probe	Anchor unit	Pairs	RMSE (ms)
Ferrero	R1	imec0	Unit 358, 215 ms	276	10.75
Ferrero	R1	imec1	Unit 582, 305 ms	308	18.42
Ferrero	R4	imec0	Unit 515, 175 ms	115	13.57
Ferrero	R4	imec1	Unit 434, 175 ms	123	18.66
Milka	R1	imec1	Unit 27, 25 ms	127	16.44
Milka	R4	imec0	Unit 102, 55 ms	210	6.12
Milka	R4	imec1	Unit 283, -95 ms	211	8.77
Milka	R2	imec0	3-unit fusion	140	1.62

Tab. 3: Sessions and alignment statistics.

333 matched pairs, after correcting for a ~ 112 s constant time-base offset between the two clocks’ absolute zero points — well within the 20 ms tolerance used elsewhere, and direct confirmation that the spike-anchor method achieves real, externally-verified precision rather than only internal self-consistency.

4.3 Alignment Diagnostics

Figure 3 shows alignment diagnostics for Ferrero R1 imec0 (276 pairs, RMSE = 10.75 ms): the stretch $s \approx 1.000002$ (≈ 2 ppm clock-rate difference) is constant over the session (flat running-median residual), confirming a single linear warp suffices. A cluster of inverse-pull events (trials 50–90, mean inter-trial interval 14.7 s) identifies a period of task disengagement that should be excluded from learning-related analyses. Figure 4 shows the anchor unit itself, Unit 358, firing reliably at 215 ms post-pull (80% trial reliability); its elevated ISI violation rate (7.92%) does not affect the warp since sigma-clipping removes outlier matches before fitting.

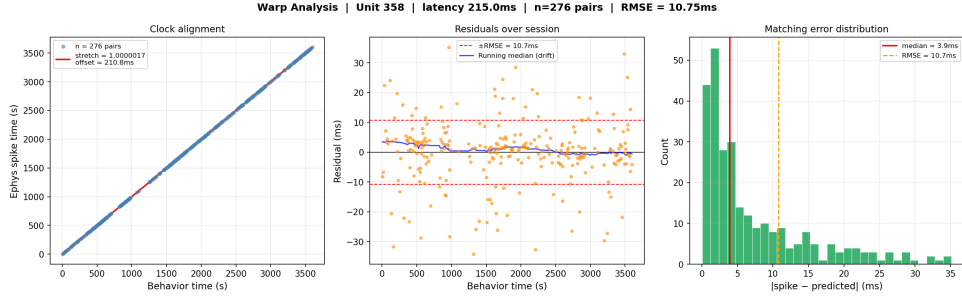


Fig. 3: Warp diagnostic, Ferrero R1 imec0. Left: Behaviour vs. ephys time, 276 matched pairs; fitted warp ($s = 1.000002$, $\delta = 211$ ms). Centre: Residuals over session, flat running median confirms no nonlinear drift. Right: Matching error distribution; median 3.9 ms, RMSE 10.75 ms.

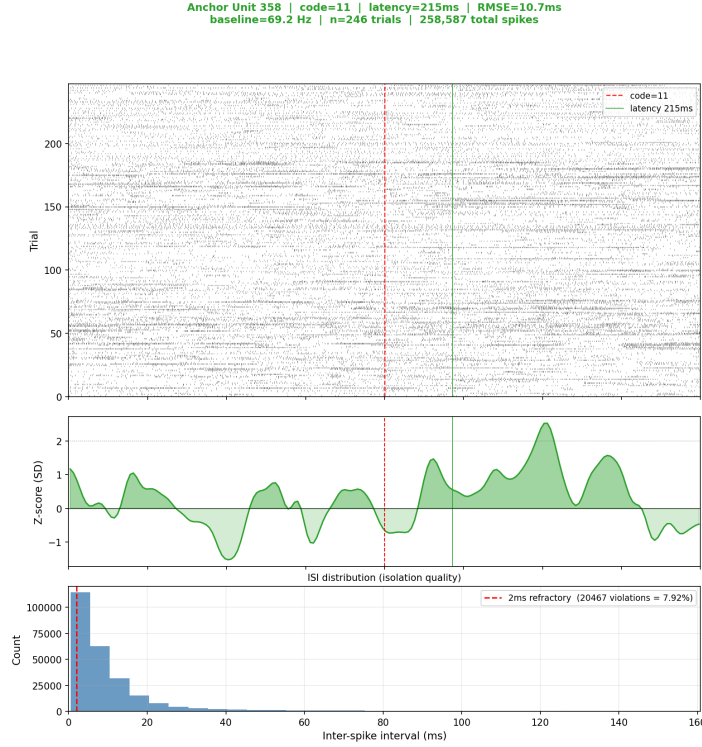


Fig. 4: Anchor Unit 358, Ferrero R1 imec0. Top: Raster across 246 pull trials, green line: 215 ms latency window. Middle: Z-scored PSTH. Bottom: ISI distribution (7.92% refractory violations).

4.4 Neural Circuit Analysis

Warped behaviour event times $t_i^{\text{ephs}} = s \cdot t_i^{\text{bhv}} + \delta$ were used to compute, for each good unit, a Z-scored PSTH (20 ms bins, ± 1 s, baseline -1 to -0.1 s). Units with $|Z| > 2.0$ were classified MOTOR (peak pre-event), REWARD (peak post-event), SUPPRESSED (trough post-event), or UNRESPONSIVE.

Across 210 good units (Ferrero R1 imec0): 46 MOTOR (21.9%), 119 REWARD (56.7%), 36 SUPPRESSED (17.1%), 9 UNRESPONSIVE (4.3%). In the 1.62 ms-RMSE session (Milka R2 imec0, 109 units): 17 Motor (15.6%), 73 Reward (67.0%), 2 Suppressed (1.8%), 17 Unresponsive (15.6%) — the higher Reward fraction likely reflects probe placement rather than alignment quality.

Figure 6 shows the population timing structure: Motor units cluster in $[-600, 0]$ ms; Reward

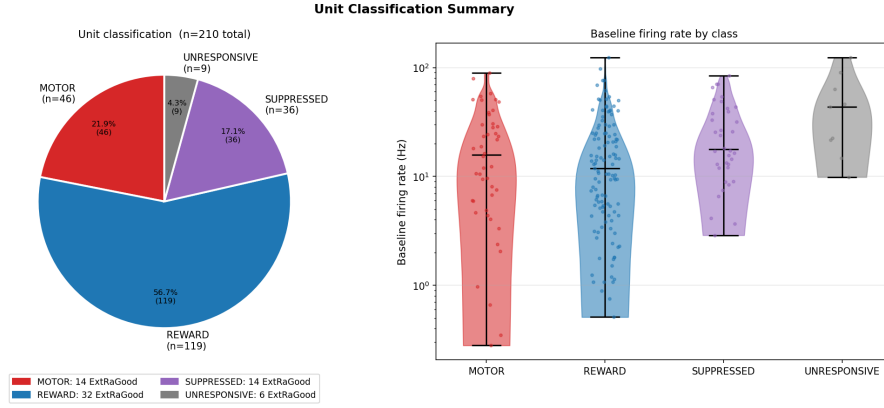


Fig. 5: Unit classification, Ferrero R1 imec0 ($n = 210$). Left: Class proportions. Right: Baseline firing rate by class (log scale); Motor and Suppressed units have higher baseline rates than Reward units, consistent with interneuron vs. output-neuron subtypes in striatum.

units span $[0, +1000]$ ms with a secondary peak near $+600$ ms, possibly corresponding to water contact and consummatory licking onset.

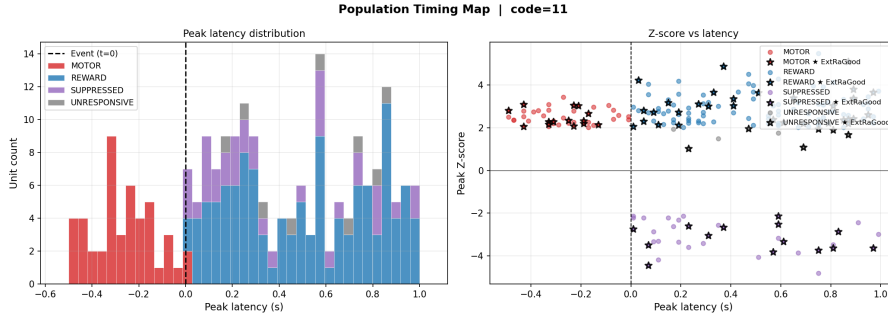


Fig. 6: Population timing map, Ferrero R1 imec0, code 11. Left: Peak latency histogram by class. Right: Peak Z-score vs. latency; ExtrRaGood units marked as stars.

Figure 8 sorts all units by peak latency, revealing a clear temporal cascade: Motor units (bottom) active before $t = 0$ and suppressed after; Reward units (middle) activating from 0 to $+1$ s; Suppressed units (top) showing sustained post-pull decrease. The 3D neural manifold (Fig. 9, PCA on the population PSTH matrix, 201 responsive units, PC1–3 cumulative 63.9% variance) shows geometrically separated pre- and post-event trajectories, with no return to baseline within $+1$ s — a sustained population state change following reward delivery.

5 Cross-Phase Discussion and Conclusion

5.1 Why Each Phase Succeeded or Failed

All three phases illustrate one principle: **alignment precision requires a signal specific to the target event, not merely correlated with it.** Video alignment succeeded qualitatively (138/138 trials) but retained second-scale imprecision, limited further by an overly narrow optimisation bound. The ephys feature phase failed outright because every one of 440+ continuous signals responded to a broad mixture of movement, licking, and arousal in addition to reward, so

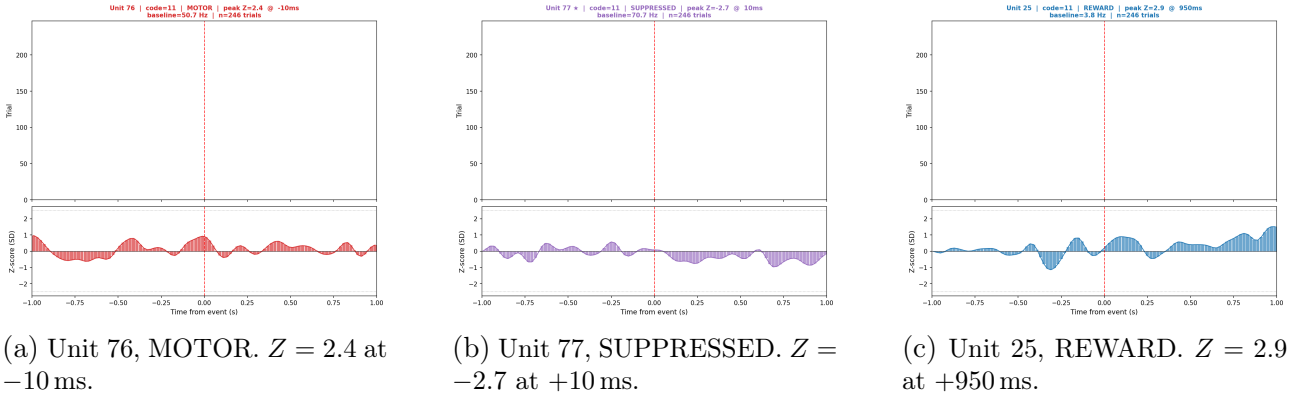


Fig. 7: Representative units, Ferrero R1 imec0, Pull (code 11). Each panel: raster (top) and Z-scored PSTH (bottom). Dashed lines: ± 2 SD classification threshold.

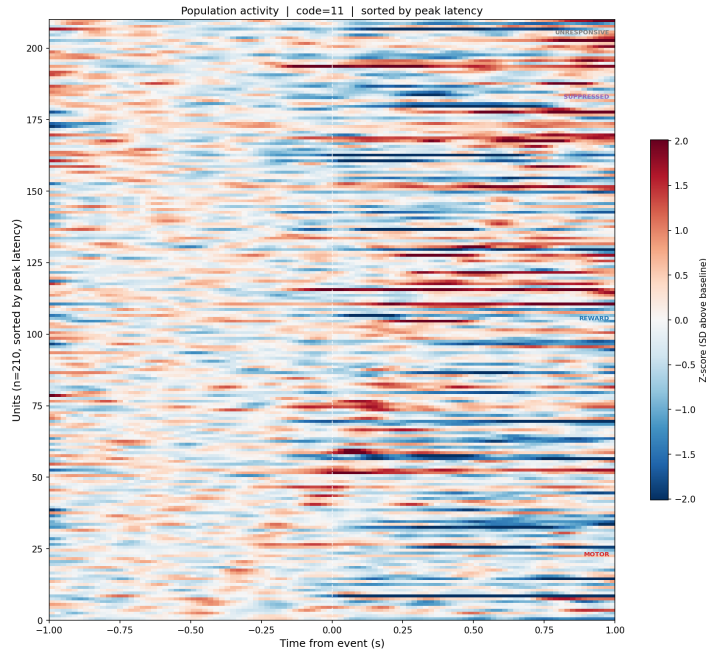


Fig. 8: Population activity heatmap, Ferrero R1 imec0 ($n = 210$, code 11). Units sorted by peak latency; colour = Z-score relative to baseline. Temporal cascade from motor preparation (pre-event) through reward processing (post-event), consistent with cortico-striatal circuit organisation.

no amount of feature engineering substituted for genuine specificity. The spike-sorted phase succeeded because individual neurons can be both highly reward-specific in their peak response *and* quiet otherwise — exactly the property a clock-anchor signal needs.

The spike-sorted approach achieved 1.6–18.7 ms RMSE across all eight probe recordings — a 10–100-fold improvement over the continuous-feature baseline, independently validated against hardware sync (Sec. 4.2), and well below the 20 ms PSTH bin size used in circuit analysis. The high responsive fraction (95.7% at $|Z| > 2.0$) reflects striatal/motor-cortical recording sites; the broad Reward latency distribution (secondary peak ~ 600 ms) suggests heterogeneous reward signals spanning action outcome, water contact, and consummatory licking, consistent with sustained post-reward striatal activity reported in prior rodent electrophysiology.

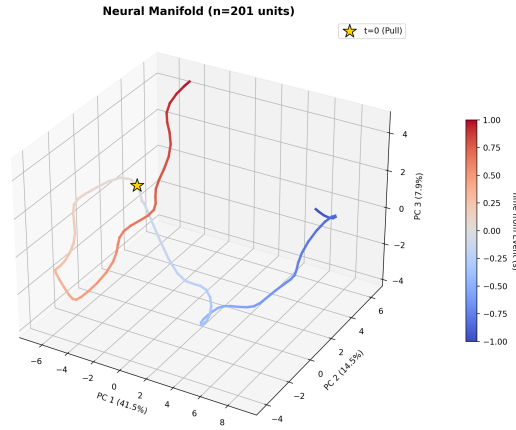


Fig. 9: Neural state-space trajectory, Ferrero R1 imec0 ($n = 201$ responsive units, code 11). First three PCs of the population PSTH matrix; colour = time from event; gold star = $t = 0$ (Pull).

Phase	Method	Typical RMSE
0	Video ensemble (PC1 + template + stability)	~ 3.7 s
1	Best continuous ephys feature (LFP deflection)	144 ms
1	All features fused	179 ms (no improvement)
2	Spike-sorted single anchor unit	1.6–18.7 ms
2	Spike-sorted multi-anchor fusion	1.62 ms (best)

Tab. 4: Alignment precision achieved by each phase’s primary method.

5.2 Limitations

- **Single-session focus.** The circuit analyses above are from one session (Ferrero R1 imec0); cross-session/cross-animal replication is required before biological conclusions can be drawn.
- **ISI violations.** Unit 358 has 7.92% refractory-period violations, suggesting possible multi-unit contamination; this does not affect alignment quality due to sigma-clipping.
- **Classification threshold.** $|Z| > 2.0$ was chosen empirically; a permutation test would be preferable for formal statistical claims.
- **Inverse-pull period.** Trials 50–90 (Ferrero R1) reflect task disengagement and should be excluded from motivationally sensitive analyses.

5.3 Conclusion

Video-based alignment (Phase 0) established that ensemble confidence-weighted detection aligns trials reliably at second-scale precision. Continuous-feature ephys alignment (Phase 1) tested 440+ engineered features and established conclusively that none were specific enough to reward delivery for precise alignment (best: 144 ms RMSE), motivating the transition to spike-sorted anchor units (Phase 2), which achieved 1.6–18.7 ms RMSE across eight probe recordings — independently validated against the hardware TTL sync channel. With alignment secured, the same spike-sorted data reveals a clear functional organisation: motor units active before

the pull, reward units responding in the 0–1 s post-pull window, and a population state-space trajectory with geometrically separated pre- and post-event regions, consistent with cortico-striatal circuit organisation during goal-directed behaviour. The pipeline is fully automated and session-agnostic, and is ready for application across all eight sessions and both animals; the next phase will examine how Motor/Reward cell proportions and latency distributions vary across sessions (learning) and probe locations (anatomy).